STUDYNEST | CLASS 11 PHYSICS

MARKING SCHEME AND MODEL ANSWERS

Chapter 4: Laws of Motion

Maximum Marks: 40 | Suggested time: 60 minutes

Use after attempting the paper. Accept equivalent correct methods.

1 mark: Correct fact, formula, unit or direction.
2 marks: Principle/formula (1); explanation, working or result (1).
3 marks: Relation (1); working/reasoning (1); final result and unit (1).
4 marks: Principle (1); setup/equation (1); working (1); result/interpretation (1).
Mark splits are a guide; award partial credit for valid steps.

Mass.

F=ma.

No.

p=mv.

Zero.

Opposes relative sliding.

Inward resultant of actual forces.

mu_sN.

4 m/s^2.

-3 kg m/s.

98 N.

24 N.

10 N s.

Water exerts a forward reaction.

4 m/s^2.

-60 N.

5 m/s^2; 86.6 N.

20 N inward.

a=5 m/s^2; v=15 m/s.

Deltap=-10 N s; force=-100 N.